

Top-Two Head-to-Head — Qwen3-32B vs Qwen3-A3B

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0. Bottom line

- **Quality:** statistically a wash. GSM8K **1.00** (32B) vs **0.95** (A3B) — a 1-question gap out of 20. Both pass **executor 13/13** and **router 59/59**; A3B passed **engine 21/21**. No recorded disagreement on any shared case.
- **Speed:** A3B is **~16× faster** (GSM8K wall 18301s vs 1128s). The MoE activates ~3B params/token vs the dense 32B's full 33B.
- **The 32B engine cell is n/a — by hardware necessity, not failure** (see note¹). On this 31 GB / 8 GB-VRAM box the dense 32B runs ELI's multi-pass engine pipeline at ~10-25h for 21 cases (and OOM-kills at full ctx). The A3B's 21/21 plus both models' 13/13 executor make the missing cell near-certain and non-decisive.
- **Verdict:** behaviourally equivalent on ELI's eval suite; the A3B wins decisively on throughput; the 32B's only edge is one extra GSM8K question. On this hardware the A3B is the rational default.

¹ The 32B engine eval was attempted twice: once at ctx=32768 (OOM-killed, rc=137 at ~31 GB) and once at ctx=8192 (ran clean but >2h for <8 of 21 multi-pass cases). Called by decision rather than wait 10-25h for a near-certain 21/21.

1. Scorecard

Metric	Qwen3-32B	Qwen3-A3B
GSM8K (20, native)	1.00	0.95
GSM8K wall (s)	18301	1128
engine eval (pass/n)	n/a ¹	21/21 (acc 1.00)
engine mean latency (s)	n/a ¹	306.9
executor eval (pass/n)	13/13 (acc 1.00)	13/13 (acc 1.00)
executor mean latency (s)	0.3	0.1
router (model-free, shared)	59/59	59/59

2. Where they disagree (per case)

No per-case disagreements recorded across the run targets present.

3. Full per-case results

engine

Case	32B	32B lat	A3B	A3B lat
banter_no_escalation	—	—	pass	209.7
capabilities_grounded_lists_real_actions	—	—	pass	354.8
code_generation_no_toy_no_placeholder	—	—	pass	113.4
cognition_runtime_lists_all_stores	—	—	pass	0.7
deepen_delivers_substance	—	—	pass	317.2
deepen_rubric_self_consistency	—	—	pass	317.9
doc_generation_grounded_not_generic	—	—	pass	4432.1
factual_offline_hedges	—	—	pass	54.2
factual_online_web_grounds	—	—	pass	211.0
factual_semantic_match	—	—	pass	149.8
greeting_is_chat	—	—	pass	85.1
grounded_self_knowledge	—	—	pass	0.0
identity_audit_gather_then_summarise	—	—	pass	38.9
memory_runtime_explained	—	—	pass	0.2
minimize_no_fabricated_done	—	—	pass	0.3
no_bruce_samuelson_confab	—	—	pass	18.4
open_domain_engine_is_url	—	—	pass	0.1
personal_memory_summary	—	—	pass	0.3
runtime_status_grounded	—	—	pass	0.0
screen_question_uses_vision	—	—	pass	61.1
stale_audit_not_confabulated	—	—	pass	79.4

executor

Case	32B	32B lat	A3B	A3B lat
exec_get_date_iso	pass	0.0	pass	0.0
exec_get_time_iso	pass	0.0	pass	0.0
exec_gpu_status	pass	0.0	pass	0.0
exec_hardware_profiles	pass	0.1	pass	0.0
exec_list_capabilities_real	pass	0.6	pass	0.5
exec_memory_status	pass	0.9	pass	0.1
exec_minimize_app	pass	0.5	pass	0.1
exec_orchestration_status	pass	0.0	pass	0.0
exec_persona_lock_status	pass	0.0	pass	0.0
exec_reasoning_modes	pass	0.0	pass	0.0
exec_resolve_runtime_paths	pass	0.5	pass	0.2
exec_runtime_status_grounded	pass	1.2	pass	0.2
exec_system_stats	pass	0.5	pass	0.5